METHANOL CONVERTER VER 1

Operator Training Simulator · R-201 quench converter and synthesis loop

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PRE-COMMISSIONING AND COMMISSIONING PLAN

FAT/SAT, mechanical completion, pressure tests, catalyst, loop checks, commissioning steps

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DOCUMENT TITLE : Pre-commissioning and Commissioning Plan

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Teaching document generated from the Methanol Converter Ver 1 simulator model; numbers come from the model and the OTS tabs. Not for construction.

Rev.	Date	Revision status	Originator	Checker	Approver
0	27-Oct-26	Issued for teaching	A. Soottitantawat		

1 Purpose and scope

This plan takes the R-201 synthesis loop from factory acceptance of its packages to the hand-over of a running unit: the SIS-201 and DCS acceptance tests, mechanical completion, pressure and leak tests, drying, purging, catalyst loading and reduction, instrument loop checks, valve stroke tests and the commissioning steps. The steps from the first circulation onward are the cold start-up of SOP-MEOH-02 … 04 and the graded procedures of the simulator, so operators practise the commissioning sequence before it happens.

Teaching document generated from the simulator model, the EPC P&IDs, the C&E MEOH-V1-IN-CE-001, the SRS MEOH-V1-IN-SRS-001 and the SOP tab. Test factors are quoted as typical code practice (ASME VIII Div. 1, ASME B31.3, API 617, API 527, IEC 61511); every other number not from the model is labelled assumed.

2 Sequence and milestones

Milestone	Meaning	On this loop
Mechanical completion (MC)	Built, pressure-tested and inspected to the drawings; punch list A closed.	Test packs TP-02 … TP-14 signed; PSV-201 re-installed with its certificate; blinds removed; P&ID walk-down.
Ready for commissioning	Pre-commissioning done without process fluids.	Dried, O_2 < 0.5 %, N_2/He leak test at 75 bar g, loop checks, valve strokes, SIS-201 SAT, catalyst loaded and reduced.
Ready for start-up	Utilities in; the unit can take feed.	Loop at about 40 bar g, C-201 ≥ 50 % (964 kNm³/h), catalyst heated, PERM-1 … 6 met.
Performance test, handover	Guarantees demonstrated; owner takes over.	112.5 t/h methanol from 365.6 kNm³/h make-up at 75 bar g; C-201 6.53 MW.

3 SIS-201 and DCS factory and site acceptance tests

FAT at the works with an I/O simulator on the logic-solver inputs; SAT on site end to end, from the field device at its trip point through SIS-201 to XV-202 and the C-201 stop circuit. The SAT is the first proof test of each SIF (proof-test interval 1 year, SRS). Action codes as the C&E: C close, S stop, TP tripped and latched, F first-out, A activate, P permissive, RB reset blocked.

The column "Model check at issue" is the same test run on a copy of the simulator model (trip constants moved past the input, permissives broken one at a time, ESD and reset through the model SIS logic): 23 of 23 model-tested rows pass. Hardware rows are tested on the logic solver only.

Test	Cause	Test method	Expected effects (C&E)	Acceptance	Model check at issue	Pass / fail, sign
FAT-SIS-01	HS-200	Operate HS-200 at the SIS console	XV-202 C, FV-202 C, HIC-214 C, C-201 S, SIS TP, 1ST F, RED A, AUD A	All effects at once (no delay); first-out "Operator emergency shutdown"; latched	✓ XV-202 shut, FV-202 MAN 0 %, HIC-214 0 %, C-201 stopped, SIS TRIPPED; first-out "Operator emergency shutdown…" after 0.00 s; latched	
FAT-SIS-02	TSHH-2xx	Inject the input across the setting (high-high 300 °C) at the logic-solver input with an I/O simulator; hold 5 s	XV-202 C, FV-202 C, HIC-214 C, SIS TP, 1ST F, RED A, AUD A	All effects after 5 s (± one scan); first-out TSHH-2xx; latched until reset	✓ XV-202 shut, FV-202 MAN 0 %, HIC-214 0 %, C-201 running, SIS TRIPPED; first-out "TSHH-2xx any bed…" after 5.25 s; latched	
FAT-SIS-03	FSLL-201	Inject the input across the setting (low-low 868 kNm³/h) at the logic-solver input with an I/O simulator; hold 5 s	XV-202 C, FV-202 C, HIC-214 C, SIS TP, 1ST F, RED A, AUD A	All effects after 5 s (± one scan); first-out FSLL-201; latched until reset	✓ XV-202 shut, FV-202 MAN 0 %, HIC-214 0 %, C-201 running, SIS TRIPPED; first-out "FSLL-201 circulation below…" after 5.25 s; latched	
FAT-SIS-04	LSHH-204	Inject the input across the setting (high-high 90 % span) at the logic-solver input with an I/O simulator; hold 5 s	XV-202 C, FV-202 C, HIC-214 C, C-201 S, SIS TP, 1ST F, RED A, AUD A	All effects after 5 s (± one scan); first-out LSHH-204; latched until reset	✓ XV-202 shut, FV-202 MAN 0 %, HIC-214 0 %, C-201 stopped, SIS TRIPPED; first-out "LSHH-204 separator V-201…" after 5.25 s; latched	
FAT-SIS-05	PSHH-203	Inject the input across the setting (high-high 85 bar g) at the logic-solver input with an I/O simulator; hold 5 s	XV-202 C, FV-202 C, HIC-214 C, SIS TP, 1ST F, RED A, AUD A	All effects after 5 s (± one scan); first-out PSHH-203; latched until reset	✓ XV-202 shut, FV-202 MAN 0 %, HIC-214 0 %, C-201 running, SIS TRIPPED; first-out "PSHH-203 loop pressure…" after 5.25 s; latched	
FAT-SIS-06	SIS-RST	Trip on TSHH-2xx, keep the input above 300 °C, press reset; then clear the input and reset again. Repeat for FSLL-201 (does not block the reset).	SIS RB while a blocking cause is present; reset accepted when clear; FSLL-201 does not block	Reset refused with message, then accepted; state IDLE	✓ TSHH present: "Reset blocked: TSHH-2xx any bed thermoco…"; cleared: reset accepted (IDLE); FSLL-201 trip with the input still low: reset accepted (IDLE)	
FAT-SIS-07	PERM-1	From IDLE with all other permissives met, hold this one not met (C-201 running and FI-201 ≥ 50 % of design) and command "admit make-up"	XV-202 stays shut; permissive shown not met	Open command refused; XV-202 shut	✓ XV-202 stayed shut; "not met: C-201 running, circulation FI-201 ≥ 50 % of design."	
FAT-SIS-08	PERM-2	From IDLE with all other permissives met, hold this one not met (catalyst active, TIC-211 and TI-212 ≥ 200 °C) and command "admit make-up"	XV-202 stays shut; permissive shown not met	Open command refused; XV-202 shut	✓ XV-202 stayed shut; "not met: Catalyst active: bed 1 inlet TIC-211 and TI-212 ≥ 200 °C."	
FAT-SIS-09	PERM-3	From IDLE with all other permissives met, hold this one not met (no bed thermocouple above 280 °C) and command "admit make-up"	XV-202 stays shut; permissive shown not met	Open command refused; XV-202 shut	✓ XV-202 stayed shut; "not met: No bed thermocouple above 280 °C."	

Test	Cause	Test method	Expected effects (C&E)	Acceptance	Model check at issue	Pass / fail, sign
FAT-SIS-10	PERM-4	From IDLE with all other permissives met, hold this one not met (V-201 level below 70 % (LIC-204)) and command "admit make-up"	XV-202 stays shut; permissive shown not met	Open command refused; XV-202 shut	✓ XV-202 stayed shut; "not met: V-201 level below 70 % (LIC-204)."	
FAT-SIS-11	PERM-5	From IDLE with all other permissives met, hold this one not met (blowdown HV-205 closed) and command "admit make-up"	XV-202 stays shut; permissive shown not met	Open command refused; XV-202 shut	✓ XV-202 stayed shut; "not met: Blowdown HV-205 closed."	
FAT-SIS-12	PERM-6	From IDLE with all other permissives met, hold this one not met (FIC-202 in MAN with output ≤ 25 %) and command "admit make-up"	XV-202 stays shut; permissive shown not met	Open command refused; XV-202 shut	✓ XV-202 stayed shut; "not met: FIC-202 in MAN with output ≤ 25 %."	
FAT-SIS-13	PERM-1 … 6 all met	All permissives met; command "admit make-up"	XV-202 opens; SIS state RUN	XV-202 open, RUN	✓ XV-202 opened; SIS RUN	
FAT-SIS-14	Loss of logic-solver power	Loss of logic-solver power	XV-202 shuts (de-energise to trip); C-201 not stopped by the SIS	As expected; not in the model (logic-solver hardware)	—	
FAT-SIS-15	Loss of one redundant power supply or I/O card	Loss of one redundant power supply or I/O card	Diagnostic alarm on the DCS; no trip	As expected; not in the model (logic-solver hardware)	—	
FAT-SIS-16	Maintenance bypass of one input	Maintenance bypass of one input	Bypass alarm on the DCS; time limit; other causes still trip	As expected; not in the model (logic-solver hardware)	—	
FAT-SIS-17	Two causes in sequence	Two causes in sequence	First-out shows the first only	As expected; not in the model (logic-solver hardware)	—	
FAT-SIS-18	DCS communication loss	DCS communication loss	SIS keeps running; communication alarm on the DCS	As expected; not in the model (logic-solver hardware)	—	
FAT-SIS-19	Response time	Response time	Scan 0.25 s + 5 s delay; logic-solver part of the 7.75 s budget (SRS)	As expected; not in the model (logic-solver hardware)	—	

Test	Cause	Test method	Expected effects (C&E)	Acceptance	Model check at issue	Pass / fail, sign
FAT-DCS-01	FIC-201 low; FIC-201 high; FIC-202 low; FIC-202 high; PIC-203 low; PIC-203 high; LIC-204 low; LIC-204 high; TIC-211 low; TIC-211 high; TIC-221 low; TIC-221 high; TIC-231 low; TIC-231 high; TIC-241 low;	Inject each alarm input across its setting in the DCS FAT	Amber alarm and horn at the C&E setting, suppression per C&E notes 7–8	Every alarm at its C&E setting	✓ 26 of 26 C&E alarm settings equal the alarm list ALM; 1 event alarm (no setting) tested by its event	

Test	Cause	Test method	Expected effects (C&E)	Acceptance	Model check at issue	Pass / fail, sign
	TIC-241 high; TI-214 high; TI-224 high; TI-234 high; TI-244 high; TI-253 high; AI-207 high; AI-208 low; FI-203 high; PDI-235 high; XI-210 low; C-201 stopped					
FAT-DCS-02	FIC-201	Configuration check of FIC-201: Converter feed (circulation) → C-201 speed	Range 0–2600 kNm³/h, reverse acting, initial tuning Kc 0.4, Ti 6	Matches the instrument list and the simulator	✓ matches makeCtrls()	
FAT-DCS-03	FIC-202	Configuration check of FIC-202: Make-up syngas → FV-202 (fail closed)	Range 0–600 kNm³/h, reverse acting, initial tuning Kc 0.3, Ti 5	Matches the instrument list and the simulator	✓ matches makeCtrls()	
FAT-DCS-04	PIC-203	Configuration check of PIC-203: Loop pressure at V-201 → purge PV-203 (fail closed)	Range 0–100 bar g, direct acting, initial tuning Kc 3, Ti 60	Matches the instrument list and the simulator	✓ matches makeCtrls()	
FAT-DCS-05	LIC-204	Configuration check of LIC-204: V-201 crude methanol level → LV-204 (fail closed)	Range 0–100 %, direct acting, initial tuning Kc 1.5, Ti 120	Matches the instrument list and the simulator	✓ matches makeCtrls()	
FAT-DCS-06	TIC-211	Configuration check of TIC-211: Bed 1 inlet → TV-211 E-201 cold-side bypass (fail open)	Range 100–300 °C, direct acting, initial tuning Kc 1.2, Ti 60	Matches the instrument list and the simulator	✓ matches makeCtrls()	
FAT-DCS-07	TIC-221	Configuration check of TIC-221: Bed 2 inlet → quench TV-221 (fail open)	Range 100–300 °C, direct acting, initial tuning Kc 1, Ti 40	Matches the instrument list and the simulator	✓ matches makeCtrls()	
FAT-DCS-08	TIC-231	Configuration check of TIC-231: Bed 3 inlet → quench TV-231 (fail open)	Range 100–300 °C, direct acting, initial tuning Kc 1, Ti 40	Matches the instrument list and the simulator	✓ matches makeCtrls()	
FAT-DCS-09	TIC-241	Configuration check of TIC-241: Bed 4 inlet → quench TV-241 (fail open)	Range 100–300 °C, direct acting, initial tuning Kc 1, Ti 40	Matches the instrument list and the simulator	✓ matches makeCtrls()	
FAT-DCS-10	HIC-214	Configuration check of HIC-214: E-204 start-up heater steam valve (fail closed)	Manual loader 0–100 %	Matches the instrument list and the simulator	✓ matches makeCtrls()	

4 Mechanical completion checklist

No.	Item	Reference	Hold
MC-01	Pressure test packs TP-02 … TP-14 tested and records signed (vessels in the shop, piping in the field)	MEOH-V1-PM-COM-001 §5	★
MC-02	C-201 casing shop test certificate (API 617) received	PKG-03	
MC-03	P&ID walk-down against MEOH-PID-001 / 002; red-lines recorded	MEOH-PID-001 / 002	
MC-04	PSV-201 installed after the hydrotest with its shop set-pressure certificate (88 bar g), inlet and outlet valves car-sealed open; the approved RLB devices (PSV-205, E-203 CW-side relief, E-204 steam-side PSV) installed with their certificates	PKG-10	★
MC-05	Blind list: every test blind removed and signed off	Blind register	★
MC-06	Insulation, painting, supports, earthing and bonding complete	Construction	
MC-07	Electrical: insulation resistance, protection relay tests, C-201 motor solo run	PKG-14	
MC-08	Punch list A (items that stop start-up) closed; mechanical completion certificate signed	MC certificate	★

5 Pressure and leak tests

Design pressures and temperatures from the equipment list MEOH-V1-ME-EQL-001: Zone A — low-pressure side (V-201, C-201 suction, E-203 outlet piping): 88 bar g — max(1.10 × 78, 78 + 1.7) = 85.8 bar g, and ≥ PSHH-203 + 3 bar = 88 bar g → design = PSV-201 set; Zone B — C-201 discharge to V-201 inlet (C-201 casing, E-202 both sides, E-201 both sides, E-204 tube side, R-201, E-203 tube side): 100 bar g — ASME UG-125: every item protected by PSV-201 ≤ 110 % of its MAWP → MAWP ≥ (1.1 × 88 + 11.3) / 1.1 = 98.2 bar g, rounded up to 5 bar.

Cross-check rule (used before the list was issued): max(1.1 × operating, PSV-201 set 88 bar g + the item’s frictional pressure drop to V-201), from the model pressures (V-201 75 bar g, converter inlet 79.7 bar g, C-201 discharge = V-201 + loop ΔP 5.6 bar, bed ΔP 0.31 / 0.48 / 0.66 / 0.88 bar; exchanger split assumed); column "Rule".

Test factors (typical code practice): ASME VIII Div. 1 hydrotest 1.3 × MAWP × LSR (UG-99), pneumatic 1.1 × MAWP × LSR (UG-100); ASME B31.3 hydrotest 1.5 × design × stress ratio (345.4), pneumatic 1.1 × (345.5); compressor casing 1.5 × MAWP (API 617). LSR = 1.0 assumed until the allowable stresses at test and design temperature are confirmed from the datasheets; MAWP taken equal to the design pressure.

5.1 Worked calculation

V-201 (zone A): design 88 bar g / 120 °C (MEOH-V1-ME-EQL-001 · MEOH-V1-ME-DS-002); rule cross-check max(1.1 × 75 = 82.5, 88) = 88 bar g. Hydrotest 1.3 × 88 = 114.4 bar g at the top (+0.54 bar for 5.5 m of water at the bottom); pneumatic 1.1 × 88 = 96.8 bar g.

R-201 (zone B): design 100 bar g / 335 °C (MEOH-V1-ME-EQL-001 · MEOH-V1-ME-DS-001); the rule gives only 93 bar g because it ignores C-201 at full speed while V-201 relieves. Hydrotest 1.3 × 100 = 130.0 bar g in the shop, before internals and catalyst; 16.6 m of water adds 1.63 bar. LSR 1.0 assumed; the datasheet allowable stresses govern.

Pneumatic test of the whole loop (352 m³ at 1.1 × 100 bar g): $E = p \cdot V/(\gamma-1)\cdot[1 - (p_a/p)^{((\gamma-1)/\gamma)}] \approx 7.2$ GJ ≈ 1.7 t TNT. Therefore the loop is hydrotested item by item and test pack by test pack, and the gas test is a staged leak test with an exclusion zone.

5.2 Test packs

Pack	Item / side	Op. (bar g)	Design (bar g / °C)	Source	Rule	Code	Hydro (bar g)	Pneu. (bar g)	Leak (bar g)	Note
TP-01	C-201 casing · casing	80.6	100 / 120	MEOH-V1-ME-EQL-001 · MEOH-V1-ME-DS-007	94	API 617	150.0	—	75.0	Shop hydrotest by the vendor at 1.5 × casing MAWP (API 617); not field-tested with the loop.
TP-02	E-202 cold interchanger · shell side (loop gas to converter)	80.3	100 / 170	MEOH-V1-ME-EQL-001 · MEOH-V1-ME-DS-004	94	VIII-1	130.0	110.0	75.0	
TP-03	E-201 hot interchanger · shell side (bed-1 feed)	79.9	100 / 335	MEOH-V1-ME-EQL-001 · MEOH-V1-ME-DS-003	93	VIII-1	130.0	110.0	75.0	
TP-04	E-204 start-up heater · tube side (process)	79.8	100 / 335	MEOH-V1-ME-EQL-001 · MEOH-V1-ME-DS-006	93	VIII-1	130.0	110.0	75.0	

Pack	Item / side	Op. (bar g)	Design (bar g / °C)	Source	Rule	Code	Hydro (bar g)	Pneu. (bar g)	Leak (bar g)	Note
TP-05	R-201 converter · shell	79.7	100 / 335	MEOH-V1-ME-EQL-001 · MEOH-V1-ME-DS-001	93	VIII-1	130.0	110.0	75.0	Hydrotest in the shop before the internals and catalyst; the vessel and its supports must carry the water weight. After catalyst loading: gas leak test only.
TP-06	E-201 hot interchanger · tube side (effluent)	77.4	100 / 335	MEOH-V1-ME-EQL-001 · MEOH-V1-ME-DS-003	91	VIII-1	130.0	110.0	75.0	
TP-07	E-202 cold interchanger · tube side (effluent)	76.5	100 / 280	MEOH-V1-ME-EQL-001 · MEOH-V1-ME-DS-004	90	VIII-1	130.0	110.0	75.0	
TP-08	E-203 final cooler · tube side (process)	75.7	100 / 225	MEOH-V1-ME-EQL-001 · MEOH-V1-ME-DS-005	89	VIII-1	130.0	110.0	75.0	
TP-09	V-201 HP separator · shell	75.0	88 / 120	MEOH-V1-ME-EQL-001 · MEOH-V1-ME-DS-002	88	VIII-1	114.4	96.8	75.0	
TP-10	E-203 final cooler · cooling-water shell	4.5	10 / 225	MEOH-V1-ME-EQL-001 · MEOH-V1-ME-DS-005	—	VIII-1	13.0	11.0	at op.	CW supply 4.5 bar g assumed; protected by the proposed CW-side relief (tube rupture, RLB).
TP-11	E-204 start-up heater · HP steam shell	40.0	45 / 280	MEOH-V1-ME-EQL-001 · MEOH-V1-ME-DS-006	—	VIII-1	58.5	49.5	at op.	HP steam 40 bar g (MEOH-PID-001); protected by the proposed steam-side PSV (tube rupture, RLB).
TP-12	Loop piping zone B, PG service (C-201 discharge ... E-203 inlet: DN750-PG-202, DN750-PG-201, DN600-PG-213, DN300/350-PG-222/232/242, DN900-PG-251) · piping	80.6	100 / 335	MEOH-V1-ME-EQL-001 zone B	94	B31.3	150.0	110.0	75.0	Zone B design pressure applied to the PG piping class (assumed); design temperature of the hottest item connected.
TP-15	Loop piping zone A (E-203 outlet DN900-PG-205 to V-201, V-201 vapour to C-201 suction) · piping	75.0	88 / 120	MEOH-V1-ME-EQL-001 zone A	88	B31.3	132.0	96.8	75.0	
TP-13	Make-up line DN300-SG-202 (FV-202, XV-202) · piping	86.0	95	1.1 × supply pressure (not in the EQL)	—	B31.3	142.5	104.5	at op.	Make-up supply 87 bar a at the battery limit (model K.Pmu).
TP-14	Crude let-down DN200-CM-209 downstream of LV-204 · piping	6.0	10	assumed; = proposed PSV-205 set (RLB)	—	B31.3	15.0	11.0	at op.	Let-down pressure 6 bar (model); design 10 bar g assumed = proposed PSV-205 set on the let-down drum (gas blow-by, HAZOP R02/R03).

5.2a Relief devices: bench test before installation

Device	Where	Set	Size (RLB)	Status	Test before installation
PSV-201	V-201 top to flare	set 88 bar g	N orifice (4N6), governing case S2	on the P&ID	Bench set 88 bar g and seat tightness (API 527)
HV-205 + RO	V-201 vapour to flare	manual blowdown	restriction orifice ≈ 49 mm	on the P&ID (RO size from RLB)	Stroke and seat test; orifice bore checked against the RLB
PSV-205	let-down drum downstream of LV-204 (off-sheet)	set 10 bar g (assumed)	R (6R8) orifice	proposed, HAZOP R03; not on the P&ID	Bench set and seat tightness (API 527)

Device	Where	Set	Size (RLB)	Status	Test before installation
E-203 CW-side relief (TSV-203 + tube-rupture device)	E-203 cooling-water shell	set 10 bar g (assumed)	L (4L6) or bursting disc; TSV-203 DN20 × DN25 (thermal, typical)	proposed (RLB, Safety layers tab)	Bench set and seat tightness (API 527); bursting disc: certificate of burst pressure
E-204 steam-side PSV (tag TBC)	E-204 HP steam shell	set 45 bar g (assumed)	H (2H3) orifice	proposed (RLB)	Bench set and seat tightness (API 527)

5.3 Leak test

N_2 with 1–10 % helium tracer (assumed), raised in steps of 10 %, 50 % and 100 % to the normal loop pressure 75 bar g (below 0.9 × PSV set = 79.2 bar g, so PSV-201 does not simmer). Every flange, instrument connection and gland is sniffed; acceptance: no detectable leak and a 1 h pressure hold with no decay after temperature correction (typical). PSV-201 is removed or its inlet blinded during every hydrotest above its set pressure and re-installed with its shop certificate (API 527). R-201 is leak-tested again after catalyst loading; it is never hydrotested with catalyst inside.

6 Drying, purging, catalyst loading and reduction

Step	Activity	Acceptance / note
6.1	Drain and dry the loop after the hydrotests with hot N_2 (or dry air before any catalyst is loaded).	Dew point −20 °C or lower at the low points (assumed); liquid water damages Cu/ZnO catalyst.
6.2	Blow the HP steam line to E-204 clean; flush the E-203 cooling-water side.	Target plates clean.
6.3	Purge with N_2 by pressure cycling.	O_2 below 0.5 % at every sample point (SOP-MEOH-01).
6.4	Load the catalyst bed by bed: inert balls on the support grids, catalyst to the drawing level (dense or sock loading). Beds 1–4: 25 / 41.6 / 58.3 / 83.3 t (model).	Outage, density and level survey per bed — hold point (PKG-02).
6.5	Fit the multipoint thermocouples TE-212 … TE-244; check the quench distributors; close the manways; leak-test R-201 again.	Continuity of all 12 points; no leak.
6.6	Reduction (typical outline; the supplier's procedure governs): N_2 circulation with C-201 at reduced pressure within the datasheet N_2 case; heat with E-204 to about 150–170 °C; H_2 from about 0.5–1 % in N_2, raised in steps only when the exotherm across each bed has died down; final hold about 210–230 °C.	Water drained at V-201 is the progress measure: theory 208.3 t × 60 % CuO (assumed) / 79.55 × 18.02 ≈ 28 t. Reduced Cu is pyrophoric: keep under N_2 or loop gas.

7 Instrument loop checks and valve stroke tests

7.1 Loop checks

Tag	System	Drawing	Check
AIT-205	DCS	MEOH-PID-001	Zero and span gas at the analyzer; reading on the DCS
AIT-208	DCS	MEOH-PID-001	Zero and span gas at the analyzer; reading on the DCS
FI-203	DCS	MEOH-PID-002	Inject 0-25-50-75-100 % at the transmitter; check the DCS reading and the alarms
FI-209	DCS	MEOH-PID-002	Inject 0-25-50-75-100 % at the transmitter; check the DCS reading and the alarms
FI-212	DCS	MEOH-PID-001	Inject 0-25-50-75-100 % at the transmitter; check the DCS reading and the alarms
FI-222	DCS	MEOH-PID-001	Inject 0-25-50-75-100 % at the transmitter; check the DCS reading and the alarms
FI-232	DCS	MEOH-PID-001	Inject 0-25-50-75-100 % at the transmitter; check the DCS reading and the alarms
FI-242	DCS	MEOH-PID-001	Inject 0-25-50-75-100 % at the transmitter; check the DCS reading and the alarms
FIC-201	DCS	MEOH-PID-002	Inject 0-25-50-75-100 % at the transmitter; check the DCS reading and the alarms
FIC-202	DCS	MEOH-PID-002	Inject 0-25-50-75-100 % at the transmitter; check the DCS reading and the alarms
FSLL-201	SIS	MEOH-PID-002	End-to-end SAT: field device → SIS-201 → XV-202 (FAT-SIS sheet)
HS-200	SIS	SIS console (C&E I-5)	End-to-end SAT: field device → SIS-201 → XV-202 (FAT-SIS sheet)
LIC-204	DCS	MEOH-PID-002	Inject 0-25-50-75-100 % at the transmitter; check the DCS reading and the alarms
LSHH-204	SIS	MEOH-PID-002	End-to-end SAT: field device → SIS-201 → XV-202 (FAT-SIS sheet)
PDI-215	DCS	MEOH-PID-001	Inject 0-25-50-75-100 % at the transmitter

Tag	System	Drawing	Check
PDI-225	DCS	MEOH-PID-001	Inject 0-25-50-75-100 % at the transmitter
PDI-235	DCS	MEOH-PID-001	Inject 0-25-50-75-100 % at the transmitter
PDI-245	DCS	MEOH-PID-001	Inject 0-25-50-75-100 % at the transmitter
PI-201	DCS	MEOH-PID-002	Inject 0-25-50-75-100 % at the transmitter; check the DCS reading and the alarms
PIC-203	DCS	MEOH-PID-002	Inject 0-25-50-75-100 % at the transmitter; check the DCS reading and the alarms
PSHH-203	SIS	MEOH-PID-002	End-to-end SAT: field device → SIS-201 → XV-202 (FAT-SIS sheet)
TI-202	DCS	MEOH-PID-002	mV / RTD simulation at the head, 0-50-100 % on the DCS; continuity of every point
TI-203	DCS	MEOH-PID-002	mV / RTD simulation at the head, 0-50-100 % on the DCS; continuity of every point
TI-205	DCS	MEOH-PID-001	mV / RTD simulation at the head, 0-50-100 % on the DCS; continuity of every point
TI-212-214	DCS	MEOH-PID-001	mV / RTD simulation at the head, 0-50-100 % on the DCS; continuity of every point
TI-222-224	DCS	MEOH-PID-001	mV / RTD simulation at the head, 0-50-100 % on the DCS; continuity of every point
TI-232-234	DCS	MEOH-PID-001	mV / RTD simulation at the head, 0-50-100 % on the DCS; continuity of every point
TI-242-244	DCS	MEOH-PID-001	mV / RTD simulation at the head, 0-50-100 % on the DCS; continuity of every point
TI-251	DCS	MEOH-PID-001	mV / RTD simulation at the head, 0-50-100 % on the DCS; continuity of every point
TI-252	DCS	MEOH-PID-002	mV / RTD simulation at the head, 0-50-100 % on the DCS; continuity of every point
TI-253	DCS	MEOH-PID-002	mV / RTD simulation at the head, 0-50-100 % on the DCS; continuity of every point
TI-254	DCS	MEOH-PID-002	mV / RTD simulation at the head, 0-50-100 % on the DCS; continuity of every point
TIC-211	DCS	MEOH-PID-001	Inject 0-25-50-75-100 % at the transmitter; check the DCS reading and the alarms
TIC-221	DCS	MEOH-PID-001	Inject 0-25-50-75-100 % at the transmitter; check the DCS reading and the alarms
TIC-231	DCS	MEOH-PID-001	Inject 0-25-50-75-100 % at the transmitter; check the DCS reading and the alarms
TIC-241	DCS	MEOH-PID-001	Inject 0-25-50-75-100 % at the transmitter; check the DCS reading and the alarms
TSHH-2xx	SIS	MEOH-PID-001	End-to-end SAT: field device → SIS-201 → XV-202 (FAT-SIS sheet)

7.2 Valve stroke tests and fail positions

Stroke 0–25–50–75–100 % from the DCS, record travel and stroke time, then remove the instrument air (and, for XV-202, the solenoid power) to prove the fail position. Fail positions agree between the P&ID loop schedule and the simulator controller definitions (makeCtrls).

Valve	Service	Line	Fail (P&ID)	Fail (model)	Stroke time (acceptance, assumed)
FV-202	Make-up syngas → FV-202	DN300-SG-202	FC	FC	≤ 10 s
PV-203	Loop pressure at V-201 → purge PV-203	DN150-PU-203	FC	FC	≤ 10 s
LV-204	V-201 crude methanol level → LV-204	DN200-CM-209	FC	FC	≤ 15 s
TV-211	Bed 1 inlet → TV-211 E-201 cold-side bypass	E-201 cold-side bypass (DN600-PG-213 circuit, line no. TBC)	FO	FO	≤ 10 s
TV-221	Bed 2 inlet → quench TV-221	DN300-PG-222	FO	FO	≤ 10 s
TV-231	Bed 3 inlet → quench TV-231	DN350-PG-232	FO	FO	≤ 10 s
TV-241	Bed 4 inlet → quench TV-241	DN350-PG-242	FO	FO	≤ 10 s
HV-214	E-204 start-up heater steam valve	HP steam to E-204 (line no. TBC)	FC	FC	≤ 10 s

Valve	Service	Line	Fail (P&ID)	Fail (model)	Stroke time (acceptance, assumed)
HV-205	Manual blowdown of V-201 vapour to flare through its restriction orifice (start permissive PERM-5: closed)	V-201 vapour to flare (line no. TBC)	—	—	full open/close by hand; seat test; position switch to PERM-5
XV-202	Make-up gas shut-off, SIS final element of I-1 … I-5 (spring return, solenoid de-energises to trip)	DN300-SG-202	FC	FC	≤ 2 s (SRS response budget, assumed)

8 Commissioning steps mapped to the OTS procedures and SOPs

From CS-04 the commissioning sequence is the cold start-up: the SOPs on the SOP tab and the graded procedures on the simulator use the same steps, so each crew practises them before the first make-up gas. The emergency shutdown is proven on the plant by the SIS SAT and the HV-205 stroke test, not by a live trip at rate; crews practise it on the simulator.

Step	What is done	SOP	OTS procedure and steps
CS-01	Utilities in service: cooling water to E-203, HP steam warmed and drained up to the E-204 block valve, instrument air, nitrogen, power, flare header *MEOH-V1-EE-LL-001; MEOH-PID-001 / 002*	SOP-MEOH-01 Pre-start-up checks	—
CS-02	Loop dried, purged with N_2 (O_2 below 0.5 %), leak-tested; SIS-201 site acceptance test done with the plant stopped; all controllers in MAN at 0 %, SIS IDLE *MEOH-V1-PM-COM-001 §3–7; MEOH-V1-IN-CE-001*	SOP-MEOH-01 Pre-start-up checks	—
CS-03	Catalyst loaded bed by bed and reduced (or pre-reduced catalyst activated) to the supplier's procedure; not in the simulator *PKG-02; Transition operation tab §6*	—	—
CS-04	Loop pressured with loop gas to about 40 bar g; C-201 started and circulating at 50 % of design or more; PIC-203, LIC-204 in AUTO; quench valves shut for the heat-up *MEOH-V1-PE-HMB-001*	SOP-MEOH-02 Cold start-up through the SIS: circulation and board line-up	**Cold start-up through the SIS** 1. Start C-201 and circulate at 50 % of design or more (FI-201 ≥ 964 kNm³/h) 2. PIC-203 and LIC-204 in AUTO (the purge stays shut below 75 bar g) 3. Quench valves shut for the heat-up (TIC-221/231/241 in MAN at 5 % or less)
CS-05	Catalyst heat-up with E-204 until the start permissives are met *MEOH-V1-IN-CE-001 (PERM-1 … 6)*	SOP-MEOH-03 Catalyst heat-up with E-204	**Cold start-up through the SIS** 4. E-204 steam block valve open and HIC-214 at 50 % or more 5. TIC-211 in AUTO with its set point between 200 and 230 °C 6. Catalyst heated: TI-212 ≥ 200 °C and every bed exit ≥ 190 °C
CS-06	First make-up gas through the SIS, quench loops to AUTO, E-204 off, pressure to 75 bar g and line-out at design *MEOH-V1-PE-HMB-001 (design case)*	SOP-MEOH-04 Admitting make-up gas and line-out	**Cold start-up through the SIS** 7. Admit make-up through the SIS with FIC-202 in MAN at 25 % or less 8. Quench loops TIC-221/231/241 in AUTO at 225 / 230 / 235 °C 9. E-204 off once the reaction carries itself (HIC-214 at 0 or block valve shut) 10. Loop pressure up to 75 ±1.2 bar g 11. FIC-202 in AUTO at 366 ±5 kNm³/h and FIC-201 in AUTO at 1928 ±40 kNm³/h 12. Line out: methanol ≥ 110 t/h, no bed above 285 °C 13. Hold 10 minutes in band
CS-07	Performance test run at design (duration and guarantees by contract; 72 h assumed): methanol, make-up, purge, C-201 power, bed temperatures *MEOH-V1-PE-HMB-001*	SOP-MEOH-09 Periodic optimisation and end-of-run adjustments as the catalyst ages SOP-MEOH-10 Purge and inerts management	**Optimisation, 45 min** (graded exercise on the Exercises tab)
Step	**What is done**	**SOP**	**OTS procedure and steps**
CS-08	Planned shutdown with the catalyst kept warm (demonstrated before handover)	SOP-MEOH-05 Normal shutdown (catalyst kept warm)	**Planned shutdown** 1. Tell the front end, the distillation unit and the fuel-gas system 2. Reduce the make-up to 250 kNm³/h or less with the loops in AUTO

Step	What is done	SOP	OTS procedure and steps
			3. Stop the make-up through the SIS (normal stop, XV-202 closed) 4. Cut circulation to about 1000 kNm³/h and sweep for 5 minutes 5. Stop C-201: the loop is blocked in under pressure and the beds stay warm 6. Temperature loops TIC-211…241 in MAN 7. FIC-201 and FIC-202 in MAN at 0 %
CS-09	Hot restart from hot standby	SOP-MEOH-07 Hot restart after a trip or a short stop	**Hot restart** 1. Start C-201 and circulate at 50 % of design or more 2. PIC-203 and LIC-204 in AUTO 3. Bed 1 inlet and TI-212 at 200 °C or more (use E-204 if needed) 4. Admit make-up through the SIS with FIC-202 in MAN at 25 % or less 5. All four bed-inlet loops in AUTO at 220 / 225 / 230 / 235 °C 6. E-204 steam off 7. FIC-202 and FIC-201 in AUTO at design, loop at 75 bar g 8. Hold 10 minutes in band (methanol ≥ 110 t/h)
CS-10	Emergency shutdown and depressuring: proven on the plant by the SIS SAT and the HV-205 stroke test, practised by every crew on the simulator (not a live trip at rate) *MEOH-V1-IN-SRS-001; MEOH-V1-PE-RLB-001*	SOP-MEOH-06 Emergency shutdown and trip response	**Emergency shutdown** 1. Push the emergency shutdown (I-5) 2. Depressure the loop to flare: HV-205 open 50 % or more 3. FIC-202 and FIC-201 in MAN at 0 % 4. Loop pressure below 40 bar g
CS-11	Handover: punch list A closed, as-built P&IDs, test dossiers, SOPs approved, operators trained and graded on the OTS procedures *MEOH-V1-PM-DOC-001 (see Project & deliverables tab)*	SOP-MEOH-12 Shift hand-over and field rounds	—

9 Pre-commissioning and commissioning checklist

★ = hold point: the next phase does not start while a hold point before it is open. The tab has the same list with tick boxes and progress.

Phase	No.	Item	Reference	Hold
Mechanical completion	MC-01	Pressure test packs TP-02 … TP-14 tested and records signed (vessels in the shop, piping in the field)	MEOH-V1-PM-COM-001 §5	★
	MC-02	C-201 casing shop test certificate (API 617) received	PKG-03	
	MC-03	P&ID walk-down against MEOH-PID-001 / 002; red-lines recorded	MEOH-PID-001 / 002	
	MC-04	PSV-201 installed after the hydrotest with its shop set-pressure certificate (88 bar g), inlet and outlet valves car-sealed open; the approved RLB devices (PSV-205, E-203 CW-side relief, E-204 steam-side PSV) installed with their certificates	PKG-10	★
	MC-05	Blind list: every test blind removed and signed off	Blind register	★
	MC-06	Insulation, painting, supports, earthing and bonding complete	Construction	
	MC-07	Electrical: insulation resistance, protection relay tests, C-201 motor solo run	PKG-14	
	MC-08	Punch list A (items that stop start-up) closed; mechanical completion certificate signed	MC certificate	★
Pre-commissioning	PC-01	Lines flushed or blown; HP steam line to E-204 blown clean	Flushing plan	
	PC-02	Loop dried after the hydrotest (dew point −20 °C or lower, assumed)	COM-001 §6	★
	PC-03	N_2 purge: O_2 below 0.5 % at every sample point	SOP-MEOH-01	★
	PC-04	N_2/He leak test in steps to 75 bar g; every flange sniffed; pressure hold passed	COM-001 §5	★
	PC-05	Instrument loop checks from the instrument list (DCS and SIS loops)	MEOH-V1-IN-INL-001	
	PC-06	Control-valve and XV-202 stroke tests; fail positions on loss of air confirmed	COM-001 §7	★
	PC-07	SIS-201 SAT: every C&E cause end to end, reset and permissives (FAT-SIS sheet repeated with field devices)	MEOH-V1-IN-CE-001	★
	PC-08	C-201 lube-oil flushing, seal-gas commissioning, coupled run on N_2 at reduced pressure within the datasheet N_2 case	PKG-03	

Phase	No.	Item	Reference	Hold
	PC-09	Catalyst loaded bed by bed and witnessed; R-201 boxed up and re-tested for leaks	PKG-02	★
	PC-10	Catalyst reduced to the supplier's procedure (or pre-reduced catalyst activated)	Transition operation §6	★
Commissioning	CM-01	Utilities in service (CW, HP steam to the E-204 block valve, instrument air, N_2, power, flare)	CS-01	
	CM-02	Loop at about 40 bar g with loop gas; C-201 running ≥ 50 % of design	CS-04 · SOP-MEOH-02	
	CM-03	Catalyst heated with E-204; start permissives met	CS-05 · SOP-MEOH-03	★
	CM-04	First make-up through the SIS; line-out at design	CS-06 · SOP-MEOH-04	★
Start-up and handover	SU-01	Performance test passed	CS-07 · SOP-MEOH-09	★
	SU-02	Planned shutdown and hot restart demonstrated	CS-08, CS-09	
	SU-03	Every crew practised the emergency shutdown on the simulator	CS-10 · SOP-MEOH-06	
	SU-04	Handover dossier complete; punch list B agreed	CS-11	★

10 Engineering queries

No.	From	To	Issue	Proposed	Status
EQ-ec-01	MEOH-V1-PM-COM-001	MEOH-V1-ME-EQL-001	Test pressures are computed from design pressures derived here by the typical rule (1.1 × maximum operating, and not less than PSV-201 set 88 bar g plus the loop pressure drop from the item to V-201): 94 bar g at C-201 discharge down to 88 bar g at V-201. The equipment list is not issued yet.	Confirm the design pressures and temperatures in the EQL and the datasheets; re-issue the pressure-test table. A single loop design pressure (the highest value) would simplify the piping class.	answered by MEOH-V1-ME-EQL-001 EQL zones: A (V-201, C-201 suction, E-203 outlet piping) 88 bar g = PSV-201 set; B (C-201 casing, E-202 both sides, E-201 both sides, E-204 tube side, R-201, E-203 tube side) 100 bar g (ASME UG-125: every item protected by PSV-201 ≤ 110 % of its MAWP → MAWP ≥ (1.1 × 88 + 11.3) / 1.1 = 98.2 bar g, rounded up to 5 bar). Test packs recomputed with these values and the EQL design temperatures.
EQ-ec-02	MEOH-V1-PM-PKG-001	R-201 datasheet / model header	Catalyst quantity: 184 m³ × 1150 kg/m³ = 211.6 t (model header), but the bed masses K.Wb add up to 208.3 t (also on MEOH-PID-001). The equipment list uses a loaded bulk density of 1131 kg/m³ (184.1 m³ → 208.3 t), consistent with K.Wb; the 1150 kg/m³ in the model header is not.	Order on volume (184 m³ plus a spare, assumed 5 %) with the loaded density confirmed by the catalyst supplier; correct one of the two numbers.	open
EQ-ec-03	MEOH-V1-PM-PKG-001	C-201 datasheet (Mechanical tab)	C-201 is sized for loop gas of MW 5.83 kg/kmol. Dry-out and catalyst heat-up on nitrogen (MW 28) give about 4.8 × the gas density, so head and power at the same speed and pressure would overload the drive.	Add a nitrogen running case to the datasheet (reduced pressure about 1/4.8 of the loop pressure or reduced speed) and check the motor and drive before the PO.	open
No.	**From**	**To**	**Issue**	**Proposed**	**Status**
EQ-ec-04	MEOH-V1-PM-PKG-001	MEOH-V1-IN-SRS-001	FSLL-201 shares FT-201 with FIC-201 and TSHH-2xx uses the indicating bed thermocouples (HAZOP R01, SRS proposals FSLL-201B and TSHH 2oo3).	Decide on the SRS proposals before the R-201 requisition; include FT-201B and the dedicated thermocouple nozzles as priced	open

No.	From	To	Issue	Proposed	Status
			Extra elements change the R-201 nozzle schedule, which is frozen at the R-201 PO (critical path).	options.	
EQ-ec-05	MEOH-V1-PM-COM-001	MEOH-PID-002	The hand valves LB-204, PB-203 and BV-231 used in the SOPs and on the simulator's Field tab are not on the EPC P&IDs, so they are missing from the piping bulk and the stroke/line-up lists.	Add them to MEOH-PID-002 (LV-204 and PV-203 bypasses, BV-231) or remove them from the SOPs.	open
EQ-ec-06	MEOH-V1-PM-COM-001	MEOH-V1-PE-RLB-001	Hydrotest pressures (e.g. 114.4 bar g on V-201) exceed the PSV-201 set 88 bar g; the HV-205 restriction orifice has no size on the P&ID; E-203 CW side (10 bar g assumed) and E-204 steam side (45 bar g) are far below the loop pressure (tube-rupture case).	PSV-201 removed or blinded during the hydrotest and re-installed with its certificate; size the HV-205 orifice and settle the tube-rupture protection of E-203 and E-204 in the relief load basis before ordering PKG-05 and PKG-10.	open
EQ-ec-07	MEOH-V1-PM-COM-001	SOP tab (SOP-MEOH-01)	SOP-MEOH-01 assumes catalyst already reduced by the supplier procedure; the commissioning plan needs a decision between in-situ reduction (H_2 in N_2 circulation, low-pressure C-201 case, water drained at V-201) and pre-reduced catalyst.	Decide the catalyst form at the PKG-02 requisition; write the reduction SOP if in-situ.	open

11 Checks against the model and the other documents

Check	This document	Model or other document	Agrees
Valve fail positions: P&ID loop schedule vs model controller descriptions	FV-202 FC, PV-203 FC, LV-204 FC, TV-211 FO, TV-221 FO, TV-231 FO, TV-241 FO, HV-214 FC, HV-205 —, XV-202 FC	FV-202 FC, PV-203 FC, LV-204 FC, TV-211 FO, TV-221 FO, TV-231 FO, TV-241 FO, HV-214 FC, HV-205 —, XV-202 FC	✓
Every C&E trip and permissive is on the FAT sheet	HS-200, TSHH-2xx, FSLL-201, LSHH-204, PSHH-203, SIS-RST, PERM-1, PERM-2, PERM-3, PERM-4, PERM-5, PERM-6	HS-200, TSHH-2xx, FSLL-201, LSHH-204, PSHH-203, SIS-RST, PERM-1, PERM-2, PERM-3, PERM-4, PERM-5, PERM-6, ALARMS	✓
Every C&E alarm is on the DCS FAT	27 alarm rows	FAT-DCS-01	✓
Hydrotest pressure above design pressure (every item)	TP-01 150, TP-02 130, TP-03 130, TP-04 130, TP-05 130, TP-06 130, TP-07 130, TP-08 130, TP-09 114.4, TP-10 13, TP-11 58.5, TP-12 150, TP-15 132, TP-13 142.5, TP-14 15	design 100, 100, 100, 100, 100, 100, 100, 100, 88, 10, 45, 100, 88, 95, 10	✓
Loop design pressures ≥ PSV-201 set 88 bar g	100, 100, 100, 100, 100, 100, 100, 100, 88, 100, 88	PSV-201 88	✓
Design pressures taken from the equipment list MEOH-V1-ME-EQL-001	13 test packs	zone A 88, zone B 100 bar g	✓
EQL design pressure ≥ the typical rule (1.1 × op., PSV set + ΔP to V-201)	TP-01 100≥94, TP-02 100≥94, TP-03 100≥93, TP-04 100≥93, TP-05 100≥93, TP-06 100≥91, TP-07 100≥90, TP-08 100≥89, TP-09 88≥88, TP-12 100≥94, TP-15 88≥88	rule recomputed from the model pressures	✓
Leak-test pressure below 90 % of the PSV set	75 bar g	0.9 × 88 = 79.2	✓
PSHH-203 trip below PSV-201 set	85 bar g	88 bar g	✓
Every procedure referenced exists, step numbers within range	startup, opt, shutdown, hotstart, esd	6 OTS procedures	✓
Every SOP referenced exists in the SOP register	SOP-MEOH-01, SOP-MEOH-02, SOP-MEOH-03, SOP-MEOH-04, SOP-MEOH-09, SOP-MEOH-10, SOP-MEOH-05, SOP-MEOH-07, SOP-MEOH-06, SOP-MEOH-12	12 SOPs	✓
Catalyst in PKG-02 = Σ model bed masses	208.3 t	Σ K.Wb = 208.3 t	✓
Catalyst volume × density (model header) = Σ K.Wb (EQ-ec-02)	184 m^3 × 1150 kg/m^3 = 211.6 t	Σ K.Wb = 208.3 t	✗
SIS-201 and DCS FAT run on copies of the model	23 model-tested rows	23 pass	✓